HOW TO MAKE A RESEARCH PRESENTATION

Fall 2025

OVERVIEW

How to Write an Abstract

How to Present Your Research

Reminders

WHAT IS AN ABSTRACT?

- Summarizes larger body of writing or project
- Contains ~150-300 words
- Convince reader that your work is worth the read
- Published for scientific conferences, posters
- Can be difficult to write

COMPONENTS OF ABSTRACTS

1.Why do we care??

- What's the importance of your work and/or your motivation to do the work

~1-2 sentences

2. What is the problem?

- What you are you/lab trying to solve? Avoid too much jargon

1 sentence

COMPONENTS OF ABSTRACTS

3. How did you go about trying to solve the problem?
   - Brief overview of your methodology, variables, etc.

~2-3 sentences

4. What are your expected results
   - Did you get an answer for your problem? What was it?
   - Try not to put actual data and avoid being too vague

~1-2 sentences

COMPONENTS OF ABSTRACTS

5. What are the implications of your answer?

- Conclusions: Is it going to change the world (probably not)? Is it going to be a significant “win”? A nice hack?
- Are your results general, potentially generalizable, or specific to a particle case?
- *Your big picture might be combined with expected results*

~1-2 sentences

BE SURE TO ALSO INCLUDE

- Name:
- Author list for your poster:
- Poster Title:
- ABSTRACT

ABSTRACT EXAMPLE

Analyzing the Mass and Stoichiometry of Native Protein Complexes

Nolasco M., Fernández-Martínez J., Obado SO., Rout MP. **Laboratory of Cellular and Structural Biology, The Rockefeller University**

Nuclear Pore Complexes (NPCs) are large protein assemblies that form channels and mediate transport between the cell nucleus and its cytoplasm. Defects in the function of these complexes lead to severe diseases such as leukemia and neurodegenerative diseases. However, the exact number of proteins it takes to build each NPC subcomplex is uncertain. Through affinity purification, a method that allows us to isolate native protein complexes, in conjunction with native mass spectrometry, we aim to determine the mass and stoichiometry (copy number of each component) of some functionally relevant subcomplexes of the NPC. This information would be crucial to understanding how the NPC is built and how it functions. While our preliminary results are promising, we still have a massive knowledge gap on NPC's in most eukaryotes and inefficient methods have made identifying the composition of subcomplexes a tedious job, we are now developing a more effective approach to study the NPC in a wide array of organisms.

HOW TO PRESENT YOUR RESEARCH

Powerpoint Presentation

Speaking on it

WHY WE PRESENT OUR RESEARCH?

To communicate your progress and share what you've discovered

To receive feedback and new ideas

To practice professional communication skills

To build confidence as a researcher

Tip: You're not being graded on being perfect — your goal is to tell the story of your research.

THE BASIC FLOW

1. Introduction – What’s your topic and why does it matter?
2. Background – What’s been done before?
3. Methods – How did you do your work?
4. Results – What did you find so far?
5. Discussion – What do your findings mean?
6. Next Steps / Conclusion – What’s next? What are you learning?

Tip: Think of your slides as “signposts” guiding your audience through your project.

THINK LIKE A STORYTELLER

Every project has a beginning (question), middle (process), and end (findings).

- Hook your audience early — “We wanted to know why…”
- Use transitions: “To answer that, we did…”
- End with a strong takeaway: “This matters because…”

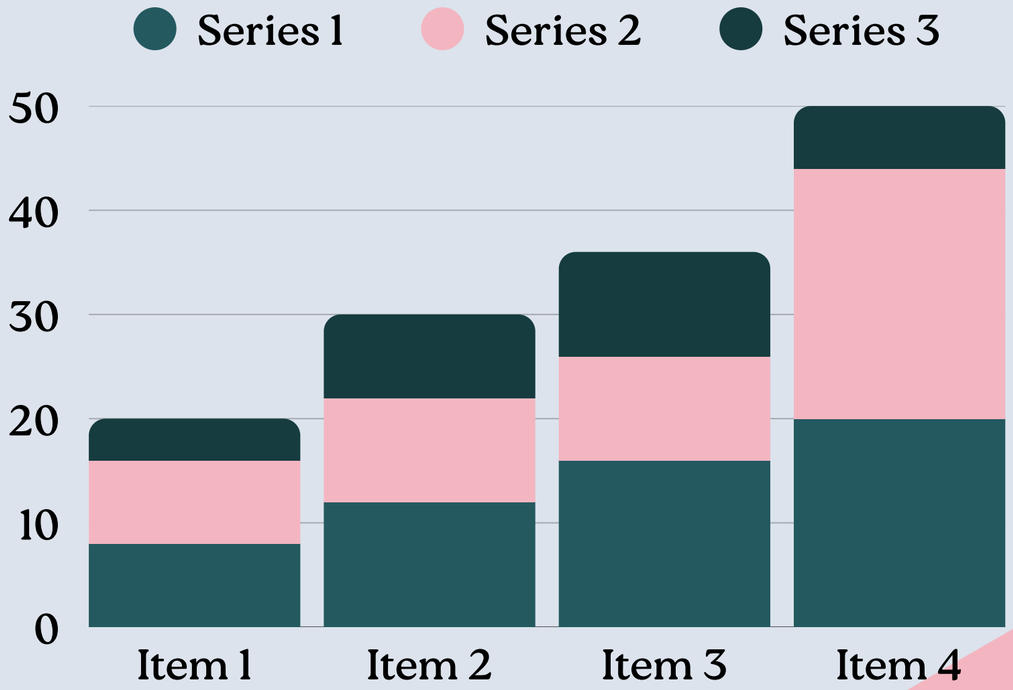
KEEP IT SIMPLE AND VISUAL

- Use minimal text (aim for under 30 words per slide)
- Choose large, readable fonts (24 pt or bigger)
- Stick to one main idea per slide
- Include images, charts, and figures where possible
- Keep backgrounds light and uncluttered

Tip: If your audience is reading, they're not listening — you are the storyteller.

VISUALIZE YOUR RESULTS

- Use graphs, tables, and visuals to summarize findings
- Label everything clearly (titles, axes, units)
- Avoid overcrowded charts — highlight key patterns
- Practice explaining each figure out loud



Example: Instead of “Raw data tables,” show a simple bar or line chart with color contrast.

HOW TO SPEAK WITH CONFIDENCE

- Know your slides — don’t memorize, internalize
- Practice transitions between topics
- Speak slowly and clearly
- Maintain eye contact and smile occasionally
- Pause for emphasis — silence can be powerful

Pro Tip: Record yourself once, it’s the fastest way to improve pacing and clarity.

STAYING ON TRACK

- Aim for 1–2 minutes per slide
- Prioritize your main points — you don’t need to say everything
- If you’re short on time, focus on results and what you’ve learned

Tip: Practice with a timer — know which slides you can skip if needed.

Q&A WITH CONFIDENCE

- Listen fully before answering
- It’s okay to say, “That’s a great question — I’ll look into that more.”
- Repeat the question before answering (helps everyone hear it)
- Keep answers short and positive

YOU’VE GOT THIS!

- Your goal is to share what you’ve learned, not to be perfect
- Your audience wants you to succeed
- Be proud of your progress — every presentation builds your skills

“Every researcher was once a student explaining their first set of results — you’re already doing the hard part.”

DISCUSSION

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Donec imperdiet nisl nec magna pellentesque, vitae eleifend odio sodales. Donec aliquet ex bibendum, pellentesque nunc sed, interdum enim.

REMINDERS!

STEM Networking-Gala

November 13th 5:30-7pm
Richard Harris Terrace
Business Casual Attire
Must RVSP to Attend!

STEP Volunteering Opportunity

College Panel Speaker: November 7th, 5-6pm

Intrepid Museum Trip: November 22nd

THANK YOU

Next Event: STEM
Networking Gala!